

Infinitesimal Calculus 2 — Lecture 3 (15.03.2026)

Limits of Multivariable Functions (cont.) & Topology of $\mathbb{R}^n$

Limits of Multivariable Functions (continued)

ε - δ Definition

$$\lim_{\bar{x} \rightarrow \bar{a}} f(\bar{x}) = L$$

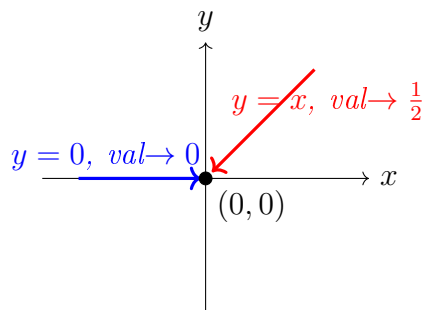
$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0$ such that:

$$0 < \|\bar{x} - \bar{a}\| < \delta \implies |f(\bar{x}) - L| < \varepsilon,$$

where $d(\bar{x}, \bar{a}) = \sqrt{(x_1 - a_1)^2 + \dots + (x_n - a_n)^2}$.

Example 1 (DNE by two paths).

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$$



- Along $y = 0$: $\frac{x \cdot 0}{x^2 + 0} = 0 \rightarrow 0$.
- Along $y = x$: $\frac{x \cdot x}{x^2 + x^2} = \frac{1}{2}$.

Two different values $\Rightarrow$ **limit does not exist**.

Example 2 (Polar coordinates — limit = 0).

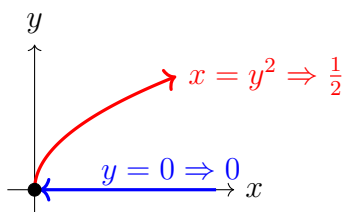
$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^2} = 0$$

Substituting $x = \rho \cos \theta$, $y = \rho \sin \theta$:

$$\frac{xy^2}{x^2 + y^2} = \rho \cos \theta \sin^2 \theta \xrightarrow{\rho \rightarrow 0} 0 \quad (\text{Squeeze: } |xy^2/(x^2 + y^2)| \leq |y| \rightarrow 0).$$

Example 3 (DNE by two paths — tricky case).

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}$$



- Along $y = 0$: limit = 0.
- Along $x = y^2$: $\frac{y^2 \cdot y^2}{y^4 + y^4} = \frac{1}{2}$.

The limit does not exist.

Example 4 (Polar with bounding trick — limit = 0).

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2 (x^2 + y^2)}{x^4 + y^4}$$

Converting to polar:

$$= \lim_{\rho \rightarrow 0} \rho^2 \cdot \frac{\cos^2 \theta \sin^2 \theta}{\cos^4 \theta + \sin^4 \theta}.$$

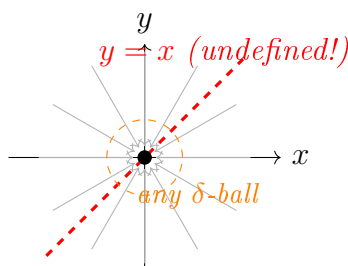
Key bound: Using $(a + b)^2 = a^2 + 2ab + b^2$ with $a = \cos^2 \theta$, $b = \sin^2 \theta$:

$$\cos^4 \theta + \sin^4 \theta = 1 - 2 \cos^2 \theta \sin^2 \theta = 1 - \frac{1}{2} \sin^2 2\theta \geq \frac{1}{2}.$$

Hence $0 \leq \frac{\cos^2 \theta \sin^2 \theta}{\cos^4 \theta + \sin^4 \theta} \leq 2$, and by Squeeze: $\rho^2 \cdot (\text{bounded}) \rightarrow 0$.

Example 5 (Not defined on a path $\Rightarrow$ limit DNE).

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x - y)}{x - y}$$



The path $y = x$ passes through $(0,0)$ and the function is undefined there ($x - y = 0$). No punctured neighbourhood of $(0,0)$ exists in which f is defined everywhere. Hence **the limit cannot be discussed / does not exist**.

Note: $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$, but $z = x - y = 0$ on the entire line $y = x$.

Continuity of Multivariable Functions

Definition 1 (Continuity at a point). $f(\bar{x})$ is **continuous** at $\bar{a}$ if:

$$\lim_{\bar{x} \rightarrow \bar{a}} f(\bar{x}) = f(\bar{a}).$$

For two variables: $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = f(x_0,y_0)$.

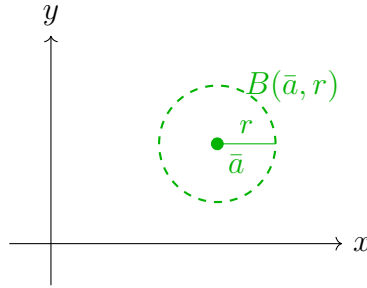
The **increment** of f at (x_0, y_0) :

$$\Delta f(x_0, y_0) = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0).$$

Continuity $\iff \lim_{(x,y) \rightarrow (x_0,y_0)} \Delta f(x_0, y_0) = 0$.

Topology of Euclidean Space $\mathbb{R}^n$

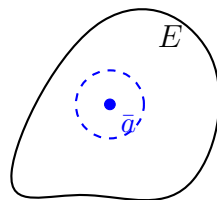
Definition 2 (Open ball). $B(\bar{a}, r) = \{\bar{x} : \|\bar{x} - \bar{a}\| < r\}$ is the **open ball** of radius r centred at $\bar{a}$.



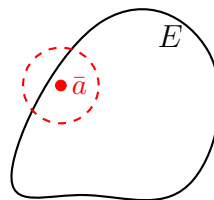
Definition 3 (Interior point (Def. 2)). $\bar{a}$ is an **interior point** of E if $\exists r > 0$ such that $B(\bar{a}, r) \subseteq E$.

Definition 4 (Exterior point (Def. 3)). $\bar{a}$ is an **exterior point** of E if $\exists r > 0$ such that $B(\bar{a}, r) \cap E = \emptyset$.

Definition 5 (Boundary point (Def. 4)). $\bar{a}$ is a **boundary point** of E if every $B(\bar{a}, r)$ meets both E and $\mathbb{R}^n \setminus E$.



Interior point
 $B(\bar{a}, r) \subseteq E$



Boundary point
Ball meets E and E^c

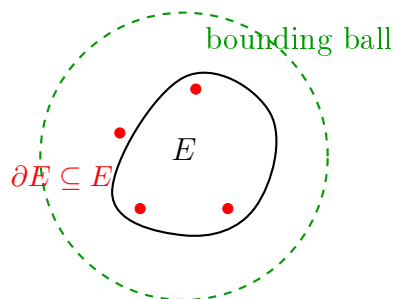
Definition 6 (Interior, Exterior, Boundary).

$$\text{Int } E, \quad \text{Ext } E, \quad \partial E.$$

Definition 7 (Open set (Def. 8)). E is **open** $\iff E = \text{Int } E$ (every point is interior).

Definition 8 (Closed set (Def. 9)). E is **closed** $\iff \partial E \subseteq E$. Remark: E closed $\iff \mathbb{R}^n \setminus E$ is open.

Definition 9 (Bounded (Def. 10) & Compact (Def. 11)). E is **bounded** if $E \subseteq B(\bar{a}, r)$ for some ball. E is **compact** if it is closed and bounded.



Compact set: closed + bounded

Weierstrass Theorems

Theorem 1 (Weierstrass 1). A function $f(\bar{x})$ continuous on a compact set is bounded on it.

Theorem 2 (Weierstrass 2). If $f(\bar{x})$ is continuous on compact E , then f attains $M = \sup_E f(\bar{x})$ and $m = \inf_E f(\bar{x})$.